TOWARDS MORE EFFECTIVE FAULT DETECTION IN LLM-BASED UNIT TEST GENERATION

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ABSTRACT

Unit tests play a vital role in uncovering potential faults in software. While tools like EvoSuite focus on maximizing code coverage, recent advances in large language models (LLMs) have shifted attention toward LLM-based test generation. However, code coverage metrics—such as line and branch coverage—remain overly emphasized in reported research, despite being weak indicators of a test suite's fault-detection capability. In contrast, *mutation score* offers a more reliable and stringent measure, as demonstrated in our findings where some test suites achieve 100% coverage but only 4% mutation score. Although a few studies consider mutation score, the effectiveness of LLMs in killing mutants remains underexplored.

In this paper, we propose MUTGEN, a mutation-guided, LLM-based test generation approach that incorporates mutation feedback directly into the prompt. Evaluated on 204 subjects from two benchmarks, MUTGEN significantly outperforms both EvoSuite and vanilla prompt-based strategies in terms of mutation score. Furthermore, MUTGEN introduces an iterative generation mechanism that pushes the limits of LLMs in killing additional mutants. Our study also provide insights into the limitations of LLM-based generation, analyzing the reasons for live and uncovered mutants, and the impact of different mutation operators on generation effectiveness.

Keywords Unit Test Generation · Mutation Testing · Large Language Models

1 Introduction

Unit test generation is a fundamental task in software engineering, crucial for ensuring code quality and reducing manual testing effort [Naik and Tripathy, 2011, Fraser and Arcuri, 2014, Corno et al., 2004, Anand et al., 2013]. With the rapid advancement of Large Language Models (LLMs), research has shifted from traditional techniques—such as random testing [Pacheco and Ernst, 2007, Hamlet, 1994, Andrews et al., 2011, Oriat, 2005, Pacheco et al., 2007] and search-based methods [Fraser and Arcuri, 2011, Rojas et al., 2016, 2015, Vivanti et al., 2013]—to LLM-based approaches for generating unit tests [Fan et al., 2023, Wang et al., 2024, Hou et al., 2024].

However, most existing work evaluates generated test cases primarily based on code coverage metrics, such as line or branch coverage. According to prior studies [Chekam et al., 2017, Andrews et al., 2006, Inozemtseva and Holmes, 2014], high code coverage does not necessarily imply strong fault detection capability. Other research [Chekam et al.,

2017, Alshahwan et al., 2024, Foster et al., 2025] suggests that mutation score is a more reliable and meaningful metric for evaluating the effectiveness of test cases. Mutation testing introduces artificial faults, known as *mutants*, into the program by applying syntactic changes using mutation operators. The generated test cases are executed against the mutants, and if a test causes a mutant to fail (i.e., "kills" it), it is considered effective. The mutation score is the ratio of killed mutants over the total number of generated mutants.

Recent efforts have begun exploring mutation-based evaluation in the context of LLM-generated test cases. For example, Foster et al. [Foster et al., 2025] investigate regression testing scenarios in privacy-critical Kotlin applications such as WhatsApp and Instagram. They focus on generating test cases using LLMs based on existing ones, and on generating mutants using LLMs, which introduces challenges such as detecting equivalent mutants that no test can kill. Another work by Dakhel et al. [Dakhel et al., 2024] targets Python benchmarks and proposes a prompt-based approach to improve mutation score. However, they iteratively add a single mutant until they cannot kill the new mutant. Therefore, they do not attempt to converge towards a maximum mutation score to maximize fault detection. Moreover, their study does not analyze the reasons why live mutants remain undetected, nor does it consider uncovered mutants, which are challenges that we address in our study.

While these efforts represent promising first steps, the use of mutation testing as a core evaluation and improvement strategy for LLM-generated test cases remains underexplored. To address this gap, we propose MUTGEN, a mutation-guided LLM-based approach that explicitly aims to maximize mutation score. Specifically, a key feature of MUTGEN is that we incorporate mutation feedback (i.e., live and uncovered mutant information) into prompt construction to guide the LLM in generating more effective test cases. Further, as discussed in prior studies [Tian et al., 2025, Wang et al., 2025a, Pan et al., 2024], incorporating a fixing step is crucial for improving test generation effectiveness. While prior work primarily focuses on improving metrics such as code coverage or test passing rate, our work explicitly targets the fault detection capability of test cases, measured by the mutation score. To this end, our approach introduces a fixing step that specifically addresses assertion failures in the generated test cases.

Moreover, we observe that comments in the target code can mislead LLMs into producing irrelevant or incorrect tests. To address this, we introduce a code summarization step during preprocessing that replaces misleading comments with concise summaries, which are then embedded into the prompt. Last, since achieving maximum mutation score is inherently difficult, we further introduce an iterative test generation strategy to push LLMs toward killing more live mutants. Once the mutation score converges, we analyze both uncovered and live mutants, as well as specific mutation operators that remain particularly challenging for the LLM.

To evaluate MUTGEN, we use two datasets: HumanEval-Java, widely adopted in prior work [Tian et al., 2025, Dakhel et al., 2024, Siddiq et al., 2024], and a new LeetCode-Java corpus containing 100 randomly selected algorithmic problems with ground-truth solutions. The experimental results show that test cases generated by MUTGEN outperform both EvoSuite, a state-of-the-art search-based technique, and a vanilla LLM-based method in terms of mutation score. MUTGEN achieves mutation scores of 89.5% and 89.1% on HumanEval-Java and LeetCode-Java, respectively, across a total of 1,144 and 1,900 mutants.

To understand the contribution of each component in MUTGEN, we conduct an ablation study using three variants: MUTGEN-S (without summarization), MUTGEN-F (without the fixing step), and MUTGEN-MF (without mutation feedback). The results confirm the importance of each component in improving the overall mutation score.

In summary, this paper makes the following contributions:

- We propose a mutation-guided, LLM-based approach (MUTGEN) for automatically generating test cases with high fault detection capability (i.e., mutation score), including an iterative generation process guided by mutation feedback to push the limits of LLMs in killing mutants.
- We evaluate our approach on a widely used benchmark dataset and a new dataset we created to increase diversity.
- We analyze the effectiveness of MUTGEN across different mutation operators and investigate the reasons behind the presence of live and uncovered mutants when MUTGEN reaches its maximum mutation score.

2 Motivating Example and Informal Description

This section goes through a simple example to informally describe our approach, MUTGEN, which we will then precisely describe and formalize in the next section. The example Java code under test shown in Figure 1, which validates a date format, includes descriptive comments and input-output examples. For clarity and to save space, we retain only the most relevant parts of the comments in the figure. Most LLM-based techniques prompt LLMs using the following standard prompt:

"Generate a JUnit test for the following method to achieve high branch and line coverage."

```java
package original;
class ValidDate {
    /** You have to write a function ......
     * The date is valid if all of the following rules are satisfied:
     * 1-3. ......
     * 4. The date should be in the format: mm-dd-yyyy
     * for example:
     * validDate('03-11-2000') => True
     * validDate('06/04/2020') => False
     * MORE EXAMPLES ...... **/
    public static Boolean validDate(String date) {
        if (date.length() != 10) {
            return false;
        }
        String[] dateArr = date.split("-");
        if (dateArr.length != 3) return false;
        int month = Integer.parseInt(dateArr[0]);
        int day = Integer.parseInt(dateArr[1]);
        int year = Integer.parseInt(dateArr[2]);
        if (month < 1 || month > 12) { return false; }
        if (month == 2) {
            if (day < 1 || day > 29) { return false; }
        } else if (month == 4 || month == 6 || month == 9 || month == 11) {
            if (day < 1 || day > 30) { return false; }
        } else {
            if (day < 1 || day > 31) { return false; }
        }
        return true;
    }
}
```

Figure 1: Example code under test

When combined with the example code (including comments), LLMs can generate test cases that achieve high line and branch coverage. However, as demonstrated in prior work [Foster et al., 2025, Chekam et al., 2017, Alshahwan et al., 2024], high coverage does not necessarily imply strong fault detection capability, for example when measured as mutation score. For instance, in our experiments, LLMs generate tests for the subject `id_81` from HumanEval-Java with 100% line and branch coverage, yet the corresponding mutation score is only 4%.

Listing 2 shows a representative test function generated by Llama-3.3. In practice, multiple such test functions may be generated, often with a variety of date inputs aimed at maximizing code coverage. However, we observe that most of these inputs fail to cover corner cases such as February 29th on leap years, likely due to the absence of such examples in the provided comment block. The comments in the target code may also introduce additional issues. For example, the comment begins with "You have to write a function…", which, when included as part of the prompt, may mislead the LLM into replicating the target code logic rather than generating a meaningful test function. Additionally, Llama-3.3 occasionally produces dates in incorrect formats such as `dd-mm-yyyy`, likely due to ambiguity or insufficient specification of the expected format in the comments provided within the program under test.

Further, when relying on mutation instead of code coverage, the generated inputs often fail to kill mutants effectively. For example, when the mutation operator changes a conditional boundary and alters line 24 to:

Listing 1: Example mutant introduced at line 24

```java
if (day <= 1 || day > 30) return false;
```

This change introduces a fault: the original code accepts day == 1 as valid, while the mutant incorrectly rejects it. We observe that LLM-generated test cases, when prompted with the original code, tend to focus on representative invalid inputs (e.g., day = 0) and valid values far from the boundary (such as day = 2 or higher). For instance, the model commonly produces a test such as:

```java
assertFalse(validDate("04-00-2025"));
```

which fails in both for the original function and mutant. However, it frequently fails to generate:

```java
assertTrue(validDate("04-01-2025"));
```

which is essential to kill the boundary-mutated version of the code.

Listing 2: Example test function generated by Llama-3.3

```
@Test
public void testDayBoundaryInApril() {
  String date = "04-01-2025";
  assertTrue(ValidDate.validDate(date));
}
```

In summary, test cases generated using the standard prompt exhibit the following limitations:

1. Comments in the target code can mislead LLMs, resulting in the generation of ineffective or irrelevant test cases.
2. Given the presence of code mutants, LLMs struggle to understand how to generate tests that kill them effectively.

To address these issues, we propose MUTGEN, an approach that improves the fault detection capability of LLM-generated test cases via a two-stage process: preprocessing and generation, which are detailed in Section 3.2 and Sections 3.3-3.5, respectively.

2.1 Preprocessing Stage

To address the first limitation above, MUTGEN removes comments from the target code and uses a prompt (shown in Figure 2) to instruct the LLM to summarize the method's purpose and input format. The prompt used is as follows:

Prompt Used for Summarizing

Summarize the following Java source code in 1-3 sentences, including its purpose and input format (if applicable).

```
{{ source code }}
```

Figure 2: Prompt used for summarizing

In this example, Llama-3.3 returns the following summary:

> *"This Java function, 'validDate', validates a given date string in the format "mm-dd-yyyy" and returns 'true' if the date is valid according to specific rules regarding month and day ranges, and 'false' otherwise. The input date string must be non-empty and follow the exact specified format. If any of these conditions are not met, the function will return 'false'."*

To address the second issue, we leverage mutation feedback to guide the LLM in generating test cases with higher fault detection capability. Before the generation stage, MUTGEN extracts mutation feedback from mutation reports produced by mutation testing tools. In this example, we use PITest [pit, Accessed: 2025] to generate a mutation report, as shown in Figure 3. From the report, MUTGEN retrieves key information, including the mutation location (line number), the mutant status (killed, survived, or uncovered), and the mutation operator applied to create the mutant. MUTGEN then integrates this feedback with the source code, forming a prompt (shown in Figure 4) that guides the LLM in generating more effective test cases during the generation stage. For example, the mutation feedback extracted from the report for the mutant shown in Listing 1 is as follows:

> *21. conditional boundary at line 24, uncovered:*
> `if (day<= 1||day>30) return false;`

This means that the existing tests fail to cover the 21st mutant, which is generated by applying a conditional boundary change at line 24, as shown on the second line.

2.2 Generation Stage

To achieve a high mutation score, MUTGEN employs a prompt augmented with mutation feedback, collected during the preprocessing stage, to guide Llama-3.3 in generating test cases, as illustrated below. For this HumanEval-Java subject, as shown in Figure 1, a single invocation of MUTGEN produces a test suite with 13 test cases, achieving a 70% mutation score across 30 mutants.

Mutations

```
12  1. negated conditional → KILLED
13  1. replaced Boolean return with True for original/ValidDate::validDate → KILLED
16  1. negated conditional → KILLED
    2. replaced Boolean return with True for original/ValidDate::validDate → KILLED
    1. negated conditional → SURVIVED Covering tests
    2. changed conditional boundary → SURVIVED Covering tests
20  3. changed conditional boundary → SURVIVED Covering tests
    4. replaced Boolean return with True for original/ValidDate::validDate → SURVIVED Covering tests
    5. negated conditional → SURVIVED Covering tests
21  1. negated conditional → SURVIVED Covering tests
    1. negated conditional → SURVIVED Covering tests
    2. changed conditional boundary → SURVIVED Covering tests
22  3. negated conditional → SURVIVED Covering tests
    4. replaced Boolean return with True for original/ValidDate::validDate → SURVIVED Covering tests
    5. changed conditional boundary → SURVIVED Covering tests
    1. negated conditional → SURVIVED Covering tests
    2. negated conditional → SURVIVED Covering tests
23  3. negated conditional → SURVIVED Covering tests
    4. negated conditional → SURVIVED Covering tests
    1. negated conditional → NO_COVERAGE
    2. changed conditional boundary → NO_COVERAGE
24  3. changed conditional boundary → NO_COVERAGE
    4. replaced Boolean return with True for original/ValidDate::validDate → NO_COVERAGE
    5. negated conditional → NO_COVERAGE
    1. changed conditional boundary → NO_COVERAGE
    2. negated conditional → NO_COVERAGE
26  3. changed conditional boundary → NO_COVERAGE
    4. replaced Boolean return with True for original/ValidDate::validDate → NO_COVERAGE
    5. negated conditional → NO_COVERAGE
28  1. replaced Boolean return with False for original/ValidDate::validDate → NO_COVERAGE
```

Figure 3: Example mutation report

Prompt Used for Generation

Unit Test Generator for Java using Junit4
Source Code Summary `{{summary}}`
Source File

`{{source code without comments}}`

`## Mutation Feedback`
Analyze the live mutant to think how to find the defects, the generated test cases must look very different with the existing ones.
...
21. conditional boundary at line 24, uncovered:
`if (day<=1||day>30) return false;`
...
Output Format in YAML
new_tests:
- test_behavior: ...
  test_name: ...
  test_code: ...
  new_imports: ...

Figure 4: Prompt used for generation

However, two of the generated test cases fail to execute due to runtime errors. To address this, MUTGEN incorporates a fixing mechanism that automatically repairs failing test cases, thereby improving both execution success rate and mutation score. In this example, MUTGEN successfully fixes one failing test case by invoking LLM with the following prompt, resulting in a 13% increase in mutation score.

To further enhance effectiveness, MUTGEN introduces an iterative generation strategy. In each iteration, additional test cases are generated to target the remaining live and uncovered mutants that were not killed by the previous test suite. After two iterations, for the running example, MUTGEN achieves a 100% mutation score. During this process, LLMs may occasionally generate test cases with method names that conflict with those already created. For example, as shown in Listing 2, the LLM generates a test case for the boundary value 04-01-2025 in April. Later, it may generate another test case for 04-31-2025 also intended to test boundary conditions. However, both may be assigned the same method name, leading to naming conflicts. To resolve this, MUTGEN incorporates an additional guideline, specifically addressing method name conflicts, into the prompt used for fixing (shown in Figure 5).

In contrast, as demonstrated in Section 4, for this subject in HumanEval-Java, using the standard prompt introduced at the beginning of this section, the LLM can only produce a test suite achieving a 53% mutation score, which remains unchanged even after four iterations.

Prompt Used for Fixing

Failed Test
The following test failed in the test suite:

```
{{ failed_test with error message }}
```

You are processing execution failures, please match only one of these guidelines and then try to correct the failed tests according to the given error messages.
1. Direct Invocation:
• Always use assertion functions directly (e.g., assertEquals(value1, value2)) instead of prefixed calls like Assert.assertEquals(value1, value2).
2. Handling assertEquals Mismatches:
• If an error follows the pattern:
• test_file_name:line_number: expected:error_value but was:correct_value
• And involves assertEquals, replace the expected value with the correct one.
• Example: assertEquals(error_value, xxx); → assertEquals(correct_value, xxx);
3. Fixing assertTrue / assertFalse Errors:
• If an error follows the pattern:
• classname:line_number:null
• And involves assertTrue or assertFalse, swap the function name while keeping all parameters unchanged.
• Example: assertTrue(xxx); → assertFalse(xxx);
4. Handling ambiguous
• If an error has the pattern:
• reference to assertEquals is ambiguous
• And involves assertEquals, add type casts to both parameters based on their respective types.
• Example: assertEquals(expectedValue, actualValue); → assertEquals((long) expectedValue, (long) actualValue);
5. Method name conflicts
• If an error follows the pattern: method is already defined in the class,
• only append a number to the method name.
• Example: public void test_null() → public void test_null1();
6. Other errors
• Try to fix the errors according to the error messages.
Output Format in YAML
The same applied to Figure 4.

Figure 5: Prompt used for fixing

3 Our Approach

In this section, we precisely describe and formalize our proposed approach, MUTGEN. As shown in Figure 6, MUTGEN takes a source program as input and returns test cases aimed at improving its fault-detection effectiveness by maximizing the test suite's mutation score. We highlight all data flows in red in the figure. Steps 1, 2, and 5 are described in Sections 3.2, 3.3, and 3.4, respectively. We do not include Section 3.5 as it reuses the components already shown in the figure. We begin with a high-level overview of MUTGEN, followed by detailed and formal descriptions of each individual component.

3.1 Overview

Let $\mathcal{M} = \{m_1, m_2, \ldots, m_n\}$ denote the set of mutants generated from the program under test. The mutation score $\mu(T)$ of a test suite T is defined as:

$$\mu(\mathcal{T}) = \frac{|\{m \in \mathcal{M} \mid \mathcal{T} \text{ kills } m\}|}{|\mathcal{M}|}$$

We aim to generate a test suite $\mathcal{T}$ such that $\mu(\mathcal{T})$ is maximized. To achieve this, we propose a two-stage process based on prompting an LLM, namely preprocessing and generation.

The preprocessing stage prepares information to guide test case generation, including code summarization and mutation feedback, as detailed in Section 3.2. In the generation stage, the LLM is prompted with mutation feedback, such as descriptions of mutation operators and diff-like representations of mutants, to generate test cases.

Let $\mathcal{R} = \{r_1, r_2, \ldots, r_k\}$ denote the set of mutation feedback entries (i.e., killed, live, uncovered), where each r_i is a tuple $\langle \text{id}_i, m_i, s_i, \text{op}_i \rangle$, respectively representing the mutant identifier, the mutated statement, the mutant status, and the applied mutation operator. MUTGEN first generates an initial test suite $\mathcal{T}$:

$$\mathcal{T} = \text{LLM}(\text{Prompt}_{\text{mut}}(\mathcal{R}))$$

Here, as further described in the following sections, $\text{Prompt}_{\text{mut}}$ is a structured prompt that encodes details of the mutation operators and specific mutant instances, as illustrated in the example $\text{Prompt}_{\text{mut}}$ shown in Figure 4.

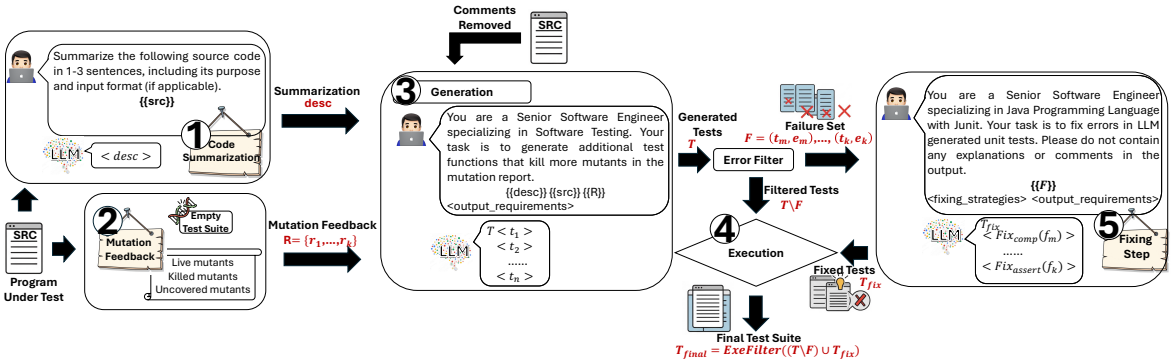


Figure 6: MUTGEN Overview

After executing $\mathcal{T}$, MUTGEN collects a set $\mathcal{F} = \{f_1, ..., f_q\}$ consisting of all test cases that encountered execution failures and corresponding error messages. For each f_i, MUTGEN constructs a second prompt $\text{Prompt}_{\text{fail}}$ (as illustrated in the example shown in Figure 5) and invokes the LLM to generate a set of fixed test cases $\mathcal{T}_{fix}$:

$$\mathcal{T}_{fix} = \{\text{LLM}(\text{Prompt}_{\text{fail}}(f_i)) \mid f_i \in \mathcal{F}\}$$

The final test suite consists of all passing test cases obtained by combining the filtered tests and the fixed tests. We denote this using the function `ExeFilter`, as follows:

$$\mathcal{T}_{\text{final}} = \texttt{ExeFilter}((\mathcal{T} \setminus \mathcal{F}) \cup \mathcal{T}_{\text{fix}})$$

This process will be detailed in Section 3.4.

Our objective is to maximize the mutation score:

$$\mathcal{T}^* = \arg\max_{\mathcal{T}} \mu(\mathcal{T})$$

Our assumption, as reported in the mutation testing literature [Chekam et al., 2017, Andrews et al., 2006], is that by maximizing the mutation score we also maximize fault detection effectiveness. In any case, achieving mutation coverage is known to be a more stringent criterion that line or branch coverage [Foster et al., 2025] and therefore enable the generation of more effective test suites. However, as revealed in recent work [Gao et al., 2025], finding an optimal prompt for LLMs remains a significant challenge and we introduce an approximate and iterative solution, detailed in Section 3.5, which we investigate empirically and whose limitations we analyze, for example in terms of mutation operators that are difficult to kill.

3.2 Preprocessing

The preprocessing stage prepares information to guide the test case generation.

Code Summarization. As illustrated in Figure 6 and the example in Section 2.1, the prompt instructs the LLM to summarize the purpose and the input format of the target code, resulting in a natural language description denoted as *desc*.

Mutation Feedback. Additionally, we apply a mutation testing tool to the code under test to generate mutation feedback. Since mutation testing tools rely on an existing test suite, we first create an empty test suite by mimicking a test function that returns directly. We then collect the mutation analysis results produced by the tool. As illustrated in Figure 6, the resulting mutation report includes information on live mutants, killed mutants, and uncovered mutants. Mutation feedback can be obtained by associating this information with the corresponding code.

As shown in the example in Section 2.1, the corresponding mutation feedback entry is represented as:

$$\begin{aligned} r_{21} = \langle & 21, \\ & if\ (day \leq 1 \| day > 30)\ return false;, \\ & \textit{uncovered}, \\ & \textit{change conditional boundary at line 24}\ \rangle \end{aligned}$$

3.3 Prompting with Mutation Feedback

We define the prompt function with mutation feedback $\text{Prompt}_{\text{mut}}$ as:

$$\text{Prompt}_{\text{mut}} : \mathcal{R} \rightarrow \text{String}$$
$$\text{Prompt}_{\text{mut}}(\mathcal{R}) = \texttt{Format}\left(desc, \bigcup_{r_i \in \mathcal{R}} r_i\right)$$

The function $\texttt{Format}(\cdot)$ organizes the given parameters into a structured prompt suitable for the LLM.

As illustrated in Figure 6 and the example in Section 2.2, the prompt constructed by MUTGEN includes a natural language code summarization ($desc$), the source code excluding comments (src), and the associated mutation feedback (R). In addition, MUTGEN explicitly specifies the expected output format from the LLM, as detailed in the prompt example Figure 4 shown in Section 2.2.

3.4 Fixing Execution Errors

After the initial generation stage, some test cases may fail due to errors in their code, MUTGEN further introduces a *fixing step* that applies a targeted prompt, as illustrated in the example prompt shown in Figure 5, to repair them.

We categorize these failures into two types:

- **Assertion Failures:** Test cases that can pass compilation but fail to execute due to violated assertions or incorrect outputs.
- **Compilation Errors:** Test cases that cannot be compiled due to syntax errors, missing types, unresolved references, or similar issues.

As shown in prior work [Yuan et al., 2024], advanced LLMs are generally capable of producing syntactically correct test cases. However, they often suffer from runtime errors, among which assertion failures are particularly common. In this paper, which targets unit testing scenarios, we focus on addressing assertion failures. To this end, MUTGEN incorporates four fix strategies into the prompt: Direct Invocation, Handling AssertEquals Mismatches, Fixing assertTrue/assertFalse Errors, and Handling Ambiguous Cases. For potential compilation errors, we employ a fifth strategy, which proactively avoids these issues during the process described in Section 3.5. For other errors, MUTGEN adopts a sixth strategy, instructing the LLM to actively repair them rather than ignoring them. All fix strategies are listed in the prompt shown in Figure 5.

We partition the set of failures into compilation and assertion failures such that

$$\mathcal{F} = \mathcal{F}_{\text{comp}} \cup \mathcal{F}_{\text{assert}}, \quad \mathcal{F}_{\text{comp}} \cap \mathcal{F}_{\text{assert}} = \emptyset.$$

Each failing test case is associated with its corresponding error message, forming a tuple $f_i = (t_i, e_i)$, where t_i is the original test case and e_i is the compiler or runtime error message.

We define a prompt construction function:

$$\text{Prompt}_{\text{fail}}(f_i) = \texttt{Format}(t_i, e_i),$$

which creates a structured natural language prompt, as illustrated in the example shown in Figure 5.

We define the specialized fix functions as:

$$\begin{aligned} \text{Fix}_{\text{comp}}(f_i) &= \text{LLM}(\text{Prompt}_{\text{fail}}(f_i)) && \text{for } f_i \in \mathcal{F}_{\text{comp}}, \\ \text{Fix}_{\text{assert}}(f_i) &= \text{LLM}(\text{Prompt}_{\text{fail}}(f_i)) && \text{for } f_i \in \mathcal{F}_{\text{assert}}. \end{aligned}$$

The outputs of these functions are the repaired test sets:

$$\begin{aligned} \mathcal{T}' &= \{\text{Fix}_{\text{comp}}(f_i) \mid f_i \in \mathcal{F}_{\text{comp}}\}, \\ \mathcal{T}'' &= \{\text{Fix}_{\text{assert}}(f_i) \mid f_i \in \mathcal{F}_{\text{assert}}\}. \end{aligned}$$

We denote the final set of fixed test cases as:

$$\mathcal{T}_{\text{fix}} = \mathcal{T}' \cup \mathcal{T}''.$$

3.5 Pushing the Limit: Iterative Generation

We propose an iterative generation process that leverages mutation feedback and failed test cases to progressively enhance the quality and effectiveness of the generated tests. At each iteration i, MUTGEN obtains a working test suite $\mathcal{T}_i$. Before proceeding to the next iteration, MUTGEN evaluates $\mathcal{T}_i$ to obtain mutation feedback $\mathcal{R}_i$. The LLM is re-invoked with $\text{Prompt}_{mut}(\mathcal{R}_i)$ to generate an initial set of new test cases. The failing cases are filtered as $\mathcal{F}_i$ and, for each $f_j \in \mathcal{F}_i$, the LLM is subsequently invoked with $\text{Prompt}_{fail}(f_j)$ to fix them. MUTGEN obtains $\Delta\mathcal{T}_i$ as the set of test cases that successfully execute. The updated test suite $\mathcal{T}_{i+1}$ is defined as:

$$\mathcal{T}_{i+1} = \mathcal{T}_i \cup \Delta\mathcal{T}_i,$$

The process continues until convergence (i.e., no significant improvement in mutation score) or until a predefined iteration limit is reached.

4 Evaluation

As previously discussed, MUTGEN is designed to enhance the fault-detection capability of LLMs in automated test generation by incorporating mutation feedback. To evaluate its effectiveness, we compare MUTGEN with two baselines: EvoSuite, the state-of-the-art search-based technique, and $\text{GEN}_{\text{vanilla}}$, a prompting-based approach that utilizes an LLM without incorporating mutation feedback and leveraging the comments provided in the program under test through summarization (as illustrated in Figure 4). We adopt Llama-3.3 70B as the base model in our evaluation due to its recent release, strong effectiveness on code-related tasks [Grattafiori et al., 2024], and open-source accessibility, which ensures reproducibility and ease of deployment in both research and practical scenarios.

In addition to effectiveness comparisons, we analyze the limitations of LLM-based test generation by examining killing ratios across different mutation operators. This allows us to identify which types of mutants are particularly challenging for LLMs to kill and offer insights into potential areas for improvement.

Finally, we conduct an ablation study to evaluate the contribution of each component within MUTGEN to the overall effectiveness.

To sum up, our evaluation addresses the following research questions.

- **RQ1:** How does MUTGEN perform in generating test cases with high fault detection capability (i.e., mutation score) compared to EvoSuite and $\text{GEN}_{\text{vanilla}}$?
- **RQ2:** Which mutation operators remain challenging for LLM-based test generation and for what reasons?
- **RQ3:** What is the individual contribution of each component of MUTGEN to the overall effectiveness?

4.1 Experiment Setup

Subjects. In our study, we primarily focus on method-level subjects, where each subject is a single Java class containing only one target method. We generate unit test cases specifically for this method. We use a widely adopted benchmark dataset that has been used in prior publications [Siddiq et al., 2024, Zheng et al., 2023, Tian et al., 2025, Islam et al.], and we also create an additional dataset to enable a more diverse evaluation. More specifically, we include the following subjects:

HumanEval-Java Following prior work [Islam et al., Siddiq et al., 2024, Tian et al., 2025, Zheng et al., 2023], we adopt the HumanEval dataset [Chen et al., 2021, Athiwaratkun et al., 2022] as one of our evaluation benchmarks. Specifically, we utilize all 160 Java subjects from the dataset. For each subject, we apply EvoSuite with its default configuration to automatically generate a test suite. We then evaluate the mutation score of the generated test suite using PITest [pit, Accessed: 2025]. Given that MUTGEN targets aims to improve mutation scores, we exclude any subjects for which the generated test suite already attains a 100% mutation score, thus obtaining 104 Java subjects for our evaluation.

Leetcode-Java We search GitHub using the keyword `"Leetcode Java"` and select the top two repositories based on the *Best match* sorting criterion. These repositories contain Java solutions to algorithmic problems from LeetCode [lee, Accessed: 2025], which are categorized into three difficulty levels: *Easy*, *Medium*, and *Hard*. We exclude all solutions corresponding to Easy problems. We collect all Java source files for the remaining *Medium* and *Hard* problems and organize them into a unified package structure following the convention used in HumanEval-Java. We use the Maven build system to compile the project and remove any subjects that failed to compile. For each valid subject, we apply

EvoSuite with its default settings to generate a test suite. Subjects for which the generated test suite achieved a 100% mutation score are once again discarded. Finally, for computational feasibility, we randomly select 50 subjects from each of the *Medium* and *Hard* sets, resulting in a total of 100 Leetcode-Java subjects used in our evaluation.

Metrics. Since our work focuses on enhancing the effectiveness of LLM-generated test cases, we adopt the **mutation score**, calculated as the ratio of killed mutants to the total number of mutants, as the primary evaluation metric. A mutant is considered *killed* if at least one generated test case causes a behavioral difference between the original program and its mutated version, typically resulting in a failed assertion or exception. In addition to the mutation score, we also report **branch coverage** and **line coverage**, in line with previous work [Alagarsamy et al., 2024, Chen et al., 2024].

Process. To address **RQ1**, we use all subjects from both datasets. As described previously, we first apply EvoSuite with its default configuration to generate test cases for each subject. Next, we generate test cases using both $\text{GEN}_{\text{vanilla}}$ and MUTGEN through a single end-to-end execution. It is important to note that, unlike prior work [Foster et al., 2025], our approach generates tests entirely from scratch, without relying on any existing test cases. We evaluate each test suite using two structural coverage metrics, **branch coverage** and **line coverage**, measured by the JaCoCo tool [jac, Accessed: 2025]. Additionally, we use PITest [pit, Accessed: 2025] to measure the **mutation score** of each test suite. To assess the statistical significance of the improvements achieved by our approach in terms of effectiveness, we performed non-parametric hypothesis testing using Vargha and Delaney's A12 statistic [Arcuri and Briand, 2011, Vargha and Delaney, 2000]. The A12 effect size was computed based on the mutation score, which ranges from 0 to 1: a value of 0.5 indicates no difference (i.e., results are due to chance), while a value greater than 0.5 suggests that MUTGEN is more likely to outperform the compared approach. Conversely, a value less than 0.5 indicates the opposite. We also conducted a paired-sample Wilcoxon signed-rank test on both coverage and mutation score to evaluate whether MUTGEN achieves statistically significant improvements on these metrics.

To address **RQ2**, we analyze the mutation operators involved in the evaluation. We first randomly select 10 subjects from each dataset and perform seven rounds of end-to-end test generation. In each round, MUTGEN generates additional test cases based on the previously generated ones. By tracking the mutation scores across iterations, we observe that the scores tend to converge after the fourth iteration. Therefore, we set the maximum number of repetitions to four in our experiments. We collect results at the fourth iteration, and analyze the distribution of mutation operators across all generated mutants and compute the *killing ratio* for each operator, that is, the proportion of killed mutants relative to the total number of mutants produced by that operator. This allows us to identify which mutation operators tend to produce mutants that are difficult or easy for LLM-generated test cases to kill. In our evaluation, mutation testing is conducted using PITest, which employs 11 mutation operators as described in [mut, Accessed: 2025a]. For mutation operators with low killing ratios, we investigate why LLMs struggle to kill these mutants. Due to space constraints, we do not present a detailed analysis for each mutation operator or for all live and uncovered mutants. Instead, we analyze a subset of cases and identify several factors contributing to the inability of LLMs to kill certain mutants. Nevertheless, our findings offer valuable insights and highlight important directions for future research in this area.

To address **RQ3**, we conduct an ablation study to evaluate the contribution of each core component in MUTGEN. Specifically, MUTGEN consists of three main components: (1) extracting and incorporating code summarization into prompts, (2) applying the fixing step to repair invalid test cases, and (3) extracting and incorporating mutation feedback into prompts. In this study, we disable one component at a time, resulting in three variants MUTGEN-S, MUTGEN-F and MUTGEN-MF. For each variant, we perform four iterations of test case generation and measure mutation score to assess the impact of each component on the effectiveness of MUTGEN.

Implementation. We describe important implementation details of MUTGEN and our evaluation setup. All subjects used in the evaluation are built using the Maven build system. We follow official guidelines to integrate EvoSuite, JaCoCo, and PITest into the Maven pipeline for automated test generation, code coverage measurement, and mutation testing, respectively. Our approach, MUTGEN, is implemented in Python and built on top of the Mutahunter framework [mut, Accessed: 2025b]. Specifically, we reuse components from Mutahunter for prompt handling, LLM interfacing, and response processing. We use Ollama [oll, Accessed: 2025] to locally deploy the Llama-3.3 70B model [Grattafiori et al., 2024] with the default configuration.

The evaluation is conducted on a Linux server equipped with a 56-core CPU, 125 GB of RAM, a single NVIDIA RTX 6000 Ada GPU, and running Ubuntu Linux 24.04.1 LTS.

4.2 Results and Analysis

Comparison with EvoSuite and $\text{GEN}_{\text{vanilla}}$ Table 1 presents the overall results of EvoSuite, $\text{GEN}_{\text{vanilla}}$, and our approach MUTGEN on the two datasets: HumanEval-Java and Leetcode-Java. The best results for each metric and dataset are highlighted.

Table 1: Summary of Results

Approach	Dataset	Line Cov	Branch Cov	Mutation Score	#Mutants per Subject	A12 effect size
EvoSuite	HumanEval-Java	95.6%	93.4%	69.5%	11	0.759
	Leetcode-Java	**99.0%**	**98.9%**	58.9%	19	0.899
GEN$_{\text{vanilla}}$	HumanEval-Java	96.2%	92.8%	77.9%	11	0.650
	Leetcode-Java	96.3%	92.7%	69.9%	19	0.734
MUTGEN	HumanEval-Java	**98.3%**	**95.8%**	**89.5%**	11	–
	Leetcode-Java	98.4%	94.8%	**89.1%**	19	–

On HumanEval-Java, for both **line coverage** and **branch coverage**, MUTGEN outperforms both EvoSuite and GEN$_{\text{vanilla}}$. On Leetcode-Java, we have the opposite since EvoSuite achieves the highest scores for coverage, surpassing both LLM-based techniques. For both datasets, differences are relatively small though, below three percentage points. Although MUTGEN is primarily designed to improve mutation score, the results also demonstrate that it consistently outperforms GEN$_{\text{vanilla}}$ in terms of line and branch coverage. This supports previous observations [Foster et al., 2025] that improvements in mutation score can also lead to enhanced code coverage.

Regarding the **mutation score**, PITest generates an average of 11 and 19 mutants per subject on HumanEval-Java and Leetcode-Java, respectively, resulting in a total of 1,144 and 1,900 mutants across the two datasets. MUTGEN achieves mutation scores of 89.5% and 89.1% on the two datasets. Compared to EvoSuite, this represents large improvements of 28.8% and 51.3%, respectively, corresponding to around 20 and 30 percentage points. Compared to GEN$_{\text{vanilla}}$, MUTGEN also significantly improves the mutation score, though to a lesser extent.

Furthermore, we report the *A12 effect size* [Arcuri and Briand, 2011] on mutation scores in the last column of Table 1. For the comparison between EvoSuite and MUTGEN, the A12 values are 0.759 and 0.899, indicating that MUTGEN has a 75.9% and 89.9% probability of achieving higher mutation scores than EvoSuite on HumanEval-Java and Leetcode-Java, respectively. Similarly, when compared with GEN$_{\text{vanilla}}$, the A12 values are 0.650 and 0.734, suggesting that MUTGEN outperforms GEN$_{\text{vanilla}}$ with 65.0% and 73.4% probabilities on the same datasets. Additionally, based on the results of the paired-sample Wilcoxon signed-rank test, we found that MUTGEN achieves statistically significant improvements ($\alpha = 0.05$) in mutation scores compared with both EvoSuite and GEN$_{\text{vanilla}}$. However, there is no statistically significant difference in coverage, as most subjects already achieve near 100% coverage across all approaches.

Overall, results suggest that MUTGEN helps significantly improve mutation scores over alternatives, with large differences, at the expense of slightly lower code coverage when compared to EvoSuite, for the most complex subject (Leetcode-Java). This may be expected as EvoSuite is designed to maximize coverage. For the other subject (HumanEval-Java), the reverse trend is however observed. Therefore, to optimize mutation scores, and therefore fault detection, we recommend the use of MUTGEN, though the impact on coverage is unclear and probably practically negligible.

Table 2: Mutant Status Counts per Operator on HumanEval-Java (Sorted by Total)

Operator	**K**	**L**	**NoCOV**	**Total**	**Killed Ratio**
NegateConditionals	398	7	3	408	97.5%
Math	209	11	3	223	93.7%
ConditionalsBoundary	186	18	4	208	89.4%
EmptyReturns	71	1	4	76	93.4%
TrueReturns	45	7	2	54	83.3%
Increments	50	2	1	53	94.3%
PrimitiveReturns	29	4	1	34	85.3%
FalseReturns	26	0	1	27	96.3%
VoidMethodCalls	18	7	1	26	69.2%
NullReturns	19	0	0	19	100.0%
InvertNegatives	16	0	0	16	100.0%
Summary	1067	57	20	1144	93.3%

Effectiveness of MUTGEN on Different Mutation Operators To evaluate the effectiveness and limitations of MUTGEN in killing mutants, we analyze the killed ratio across different mutation operators (described in [mut, Accessed: 2025a]) after running MUTGEN for four iterations. Tables 2 and 3 present the mutant status counts for each

Table 3: Mutant Status Counts per Operator on Leetcode-Java (Sorted by Total)

Operator	K	L	NoCOV	Total	Killed Ratio
Math	766	8	14	788	97.2%
NegateConditionals	474	5	14	493	96.1%
ConditionalsBoundary	288	11	1	301	95.7%
PrimitiveReturns	97	14	2	113	85.8%
VoidMethodCalls	58	2	7	67	86.6%
Increments	46	7	0	53	86.8%
EmptyReturns	24	0	2	26	92.3%
TrueReturns	15	0	11	26	57.7%
NullReturns	17	0	2	19	89.5%
FalseReturns	13	0	0	13	100.0%
InvertNegatives	1	0	1	2	50.0%
Summary	1799	47	54	1900	94.7%

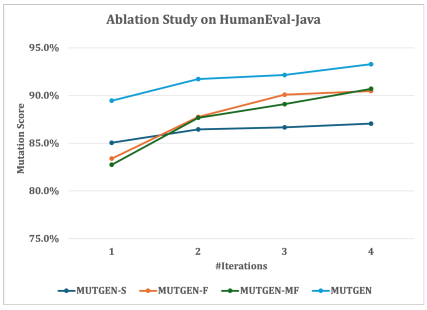
mutation operator on HumanEval-Java and Leetcode-Java, respectively. In both tables, Column **K** indicates the number of killed mutants for each corresponding mutation operator. Column **L** shows the number of live mutants (covered by not killed), while Column **NoCOV** represents the number of mutants that are not covered by any test cases. Column **Killed Ratio** denotes the proportion of killed mutants relative to the total number of mutants.

We observe that, for frequently applied mutation operators, such as `Math` and `NegateConditionals`, MUTGEN is able to progressively kill the corresponding mutants across iterations. After four iterations, MUTGEN achieves a high killed ratio. For instance, 97.2% for `Math` on Leetcode-Java. The remaining 2.8% of mutants are mostly uncovered, meaning they are not exercised by any generated test cases.

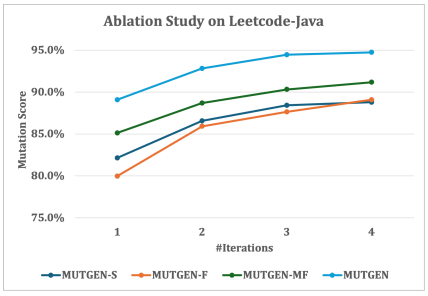
In contrast, mutation operators that appear less frequently tend to exhibit lower effectiveness. For example, `VoidMethodCalls` shows a killed ratio of 69.2% on HumanEval-Java, and `TrueReturns` achieves only 57.7% on Leetcode-Java, representing the weakest effectiveness among all mutation operators in terms of killed ratio.

For the uncovered mutants, the primary reasons are the presence of dead code or the inability of the generated test cases to reach the mutated statements, even after incorporating mutation feedback. This suggests that further improvements in mutation score may require more fine-grained information and targeted guidance. Exploring how to effectively provide such detailed feedback to LLMs remains an open direction for future research.

For live mutants, most mutation operators alter the original program logic. For example, the `MATH` operator may mutate a statement `a = b + c;` to `a = b - c;`, changing the value of the variable `a`. If `a` is not critical to the program's behavior, the LLM may fail to recognize the impact of the mutation and thus generate test cases that do not expose the fault. This observation further highlights the need for more fine-grained semantic information to guide LLMs in identifying and targeting meaningful behavioral changes, thereby improving their ability to detect live mutants.



(a) Results on HumanEval-Java



(b) Results on Leetcode-Java

Figure 7: Mutation Score Changes over 4 Iterations: Ablation Study on Both Datasets.

Ablation Study Figure 7 presents the mutation scores achieved by the ablation variants—MUTGEN-S, MUTGEN-F, MUTGEN-MF—and our approach MUTGEN across four iterations. Specifically, Figure 7(a) shows the results on

HumanEval-Java, while Figure 7(b) reports the results on Leetcode-Java. The horizontal axis represents the number of iterations, and the vertical axis denotes the mutation score.

Based on the mutation scores obtained at the fourth iteration, code summarization appears to have the most significant impact on the effectiveness of MUTGEN. As discussed in Section 2, code comments can sometimes mislead LLMs or omit essential details—such as input formats—that are crucial for guiding test case generation. The fixing stage also shows a notable impact on effectiveness, particularly on the LeetCode-Java dataset, where many generated test cases initially fail to execute successfully. Prior work has observed that LLM-generated test cases often suffer from low execution success rates. After applying fixes, however, a substantial portion of these test cases become valid and contribute to the test objectives. Our evaluation confirms that the fixing stage contributes meaningfully to improved mutation scores.

Lastly, mutation feedback also plays a role in enhancing effectiveness. As revealed in RQ2, while many mutants generated by common mutation operators are relatively easy for LLMs to kill, mutation feedback still provides effectiveness gains by guiding the model more effectively. Future work may further investigate how mutation feedback can be tailored to help LLMs kill harder-to-detect mutants, and which mutation operators benefit the most from such guidance. We leave this exploration as a direction for future research.

4.3 Threats to Validity

The threat to internal validity mainly lies in the correctness of the implementation of MUTGEN and the experimental scripts. To reduce this threat, we have carefully reviewed our code.

The threat to external validity mainly lies in the subjects and the target approaches. For subject selection, we reused benchmarks from existing publications and further enhanced subject diversity by evaluating our approach on 100 additional subjects. These were randomly sampled from a corpus of LeetCode algorithm solutions. In the future, we will evaluate MUTGEN on more complicated subjects, e.g., Defects4J. Notably, Defects4J contains real-world defects, which can provide a stronger assessment of the practical effectiveness of our approach, compared to the synthetic defects simulated via mutation testing in this study.

Regarding the baseline approaches, we selected two representative methods: (1) EvoSuite, a widely-used state-of-the-art search-based testing tool, and (2)$\text{GEN}_{\text{vanilla}}$, a vanilla LLM-based prompt approach. These baselines allow us to evaluate whether incorporating mutation feedback in MUTGEN significantly improves fault detection capability, as measured by mutation score. Although we use Llama-3.3 as the base LLM in this study, our approach is not tied to any specific model. MUTGEN is model-agnostic by design, and evaluating its performance with other LLMs is left as future work.

Several factors may affect the construct validity of our evaluation.

Randomness. Randomness may impact the effectiveness of EvoSuite, MUTGEN, and its variants. To mitigate this, and considering computational cost, we ran each tool on every subject three times and reported averaged results. We also performed statistical significance tests and calculated effect sizes, as detailed in Section 4.2.

Equivalent Mutants. Our approach uses the `PITest` mutation testing framework, which includes mechanisms to reduce the creation of equivalent mutants. While this mitigates the problem to some extent, the remaining equivalent mutants can reduce the efficiency of MUTGEN and prevent it from achieving a 100% mutation score.

Oracle Problem. We assume the original code is correct and only address assertion errors that cause Maven build failures. This simplification can lead to the generation of incorrect test oracles and result in false positives.

Data Leakage. While it is possible that some public code in benchmark datasets appears in the LLM's training data (e.g., Llama-3.3), the specific mutants we generate are unlikely to be memorized. Moreover, we compare MUTGEN with a vanilla prompting baseline, and the observed performance gains suggest that our improvements stem from prompt engineering and strategy design rather than memorization.

5 Related Work

Mutation-Guided Test Generation. Recent studies have questioned the sufficiency of coverage as an evaluation criterion. As shown in [Chekam et al., 2017, Andrews et al., 2006, Inozemtseva and Holmes, 2014], and pointed in recent work from Meta [Alshahwan et al., 2024, Foster et al., 2025], high coverage does not necessarily correlate with strong defect detection capability. Mutation testing offers a more rigorous alternative by assessing how well tests detect injected faults (mutants) [Chekam et al., 2017]. Although some prior work [Dakhel et al., 2024, Foster et al., 2025] has attempted to improve mutation scores during test generation, it does not fully explore the underlying limitations of LLMs in detecting and killing live mutants. Our work aims to address this gap by incorporating mutation feedback into

the prompt design process for LLMs, thereby enhancing the effectiveness of the generated test cases. In addition, we introduce an iterative generation process to push the limits of LLMs in killing mutants, providing insights into their current capabilities and boundaries. This exploration also highlights opportunities for future work on guiding LLMs to generate test cases with stronger defect detection capabilities.

Other techniques. Automatic unit test generation has long been a central research topic in software engineering. Traditional efforts have largely focused on search-based and random testing techniques, with tools such as EvoSuite [Fraser and Arcuri, 2011] and Randoop [Pacheco and Ernst, 2007] significantly reducing the developers' manual testing burden.

With the advent of large language models (LLMs), a new paradigm has emerged in test generation. Numerous recent surveys have summarized the rapid advancements in this area [Fan et al., 2023, Wang et al., 2024, Hou et al., 2024]; however, due to the large volume of related work, we only highlight a selection of representative examples in this section. Early research explored fine-tuning LLMs for code-related tasks. As LLMs became more powerful and general-purpose, however, prompt-based approaches gained traction, enabling test generation without task-specific training. Several studies [Siddiq et al., 2024, Yang et al., 2024, Yuan et al., 2024, Zhang et al., 2024, Li et al., 2023, Pizzorno and Berger, 2024] have investigated prompt engineering strategies and mechanisms to improve test case generation effectiveness using commercial LLMs, notably OpenAI's GPT series [Achiam et al., 2023, Ouyang et al., 2022, Brown et al., 2020].

Given the high cost of commercial LLMs, and the increasing availability of high-quality open-source alternatives such as the Llama and DeepSeek series [Grattafiori et al., 2024, Liu et al., 2024a], recent research has shifted toward optimizing prompt-based effectiveness for open-source models [Ouédraogo et al., 2024, Wang et al., 2025a, Deljouyi et al., 2024, Wang et al., 2025b, Huang et al., 2023, Kim et al., 2025]. These models are becoming popular due to their accessibility and effectiveness in code-related tasks.

LLM-based test generation has been applied across a wide range of testing scenarios. Some studies focus on multilingual test generation across languages such as Python [Wang et al., 2025a, Liu et al., 2024b, Xu et al., 2025, Tian et al., 2025], Java [Zhang et al., 2024, Wang et al., 2025a, Chen et al., 2024, Gu et al., 2024], and others [Wang et al., 2025a, Liu et al., 2024b, Zhang et al., 2025, Cheng et al., 2025, Wang et al., 2025c]. Other works target specific testing goals, such as detecting particular bugs [Liu et al., 2024b, Foster et al., 2025], API testing [Cao et al., 2024, Deng et al., 2025, Li et al., 2025, Kim et al., 2025], or generating test oracles [Khandaker et al., 2025, Dinella et al., 2022, Binta Hossain and Dwyer, 2024]. LLMs have also been integrated with traditional testing techniques, including mocking [Nan et al., 2025, ZHU et al., 2025], evolutionary algorithms [Taherkhani et al., 2024], and program analysis [Gu et al., 2025, Pan et al., 2025, Ryan et al., 2024, Pan et al., 2024]. Evaluation in these works typically emphasizes code coverage metrics, such as branch and line coverage.

Unlike prior work, our approach focuses on automatically improving the mutation score with LLMs, based on mutation testing feedback, since such score better reflects the fault-detection capability of test cases than code coverage [Chekam et al., 2017, Foster et al., 2025]. As discussed in Section 2, high code coverage does not necessarily imply strong fault detection [Chekam et al., 2017]. However, as demonstrated in our experiments (Section 4.2), improving mutation score often leads to increased code coverage as a by-product.

6 Discussion

We briefly discuss the limitations of our current work and outline directions for future research.

As described in Section 4 and 5, our experiments were conducted on independent, method-level subjects. This setup showed how well LLMs perform in killing mutants when not being confounded by broader program structures or dependencies. However, when transitioning to more complex datasets such as Defects4J, additional challenges arise, particularly in resolving dependencies to generate executable and meaningful test suites. We leave the investigation of such datasets for future work.

In addition, while we use Llama-3.3 as the base model in this study, MUTGEN is inherently model-agnostic and does not rely on any specific LLM. Though our results demonstrate the effectiveness of the approach, other models could potentially yield even better results. A more comprehensive evaluation across different LLMs is also left as future work.

Finally, although MUTGEN achieves high mutation scores on the selected subjects, there remain open challenges in pushing these scores even higher. For example, guiding LLMs to effectively kill mutants produced by certain mutation operators, such as `VoidMethodCalls`, remains difficult. In our evaluation on HumanEval-Java, this operator had only a 69.2% killing ratio. Developing strategies to better handle such challenging mutants is an important direction for future research.

7 Conclusion

In this paper, we propose MUTGEN, a mutation-guided, LLM-based approach for automatically generating unit tests with high fault detection capability. The core innovation of MUTGEN lies in incorporating mutation feedback into the prompt to help LLMs generate tests that kill more mutants. In addition, two components—code summarization and fixing—further enhance the effectiveness of the approach.

We evaluated MUTGEN on 104 subjects from the widely-used HumanEval-Java dataset and 100 subjects from Leetcode-Java, which we constructed from LeetCode algorithmic problem solutions. Experimental results show that MUTGEN significantly outperforms both EvoSuite and Llama-3.3 using a vanilla prompting strategy, achieving mutation scores of 89.5% and 89.1% on HumanEval-Java and Leetcode-Java, respectively.

Moreover, MUTGEN employs an iterative generation process to push the limits of LLMs in killing mutants. We further analyze the causes of remaining live and uncovered mutants, as well as the impact of different mutation operators, offering insights and directions for future research.

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